

Tutorial 12 Answer Key

1. $10 \pm 1\%$
2. $\frac{s+1}{s+2}$
3. $\frac{3s+2}{s-1}$
4. (a) $S_c = 3789.403 \text{ AT/wb}$, $S_{airgap} = 442097.06 \text{ AT/Wb}$
(b) 0.9 mWb
(c) 0.803 A
5. (a) 26.87 A
(b) 1 A
6. Air gap flux = 0.41 mWb
Air gap flux density = 1.03 Wb/m^2
Magnetic intensity in airgap = $0.82 \times 10^6 \text{ At/m}$
7. $S = 2 \times 10^8 \text{ AT/Wb}$, $I = 250 \text{ A}$
8. (a) 50.2 A
(b) 3.97Ω
(c) 4.16 A
9. Assuming Wall side socket voltage is 230 V then
Winding ratio = $2/23 = 0.087$
Primary side current = 0.282 A

